



Modelling Financial Markets

Chapter 12: Options and the Volatility Surface



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Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ ATM: At-The-Money
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ CRR: Cox–Ross–Rubinstein (binomial tree)
- ▣ DVOL: Deribit Volatility Index
- ▣ ETF: Exchange-Traded Fund
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GBM: Gradient Boosting Machine
- ▣ GJR-GARCH: Glosten–Jagannathan–Runkle GARCH
- ▣ GMM: Generalized Method of Moments
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ HAR-RV: Heterogeneous AutoRegressive model of Realised Volatility
- ▣ ITM: In-The-Money
- ▣ MA: Moving Average
- ▣ MZ: Mincer–Zarnowitz (regression)
- ▣ OLS: Ordinary Least Squares
- ▣ OTM: Out-of-The-Money
- ▣ QLIKE: Quasi-LIKElihood (loss function)
- ▣ RV: Realised Variance
- ▣ SSVI: Surface SVI (Gatheral–Jacquier parameterisation of the whole surface)
- ▣ SVI: Stochastic Volatility Inspired (parameterisation)
- ▣ SVIX: Simple VIX: index of simple variance swaps (Martin, 2017)
- ▣ US: United States
- ▣ UTC: Coordinated Universal Time (Temps Universel Coordonné)

Acronyms used in this material (2/2)

- ▣ **VIX: CBOE Volatility Index**
- ▣ **VIX3M: Cboe 3-Month Volatility Index**
- ▣ **VIX9D: Cboe 9-Day Volatility Index**
- ▣ **VRP: Variance Risk Premium**
- ▣ **VVIX: Cboe VIX of VIX Index**
- ▣ **XFG: Quantlet prefix of Applied Quantitative Finance (Härdle, Chen & Overbeck)**

Chapter Route and Outcomes

- An **option** gives the right, not the obligation, to buy (call) or sell (put) an asset at a fixed strike price
 - ▶ It pays only in some states: a bet on the **distribution**, not only on the direction; option prices reveal what the market expects about volatility and crashes
- Part I: pricing by replication and what a hedge leaves behind
 - ▶ Parity and static spanning; the tree and the Black–Scholes equation; the Greeks; discrete-hedging error and the gamma-weighted volatility P&L
 - ▶ Implied volatility, the smile, and a full Bitcoin surface with its risk-neutral density
- Part II: implied volatility as information
 - ▶ Variance swaps and the VIX; the variance risk premium; is the VIX a good forecast? 0DTE options (zero days to expiry)
 - ▶ Case study: are option-implied bounds on the equity premium valid and tight? AI for scientific discovery
- Outcome: estimate a surface and a density under no-arbitrage constraints, with bands, and test option-implied forecasts with the right standard errors

Chicago, 1973: Listed Options Begin



Trading floor of the Chicago Board of Trade in 2006; its members founded the Cboe in 1973

Photo: [Richie Diesterheft \(2006\)](#); CC BY 2.0; [Wikimedia](#)

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- The Chicago Board Options Exchange (Cboe) opened in 1973 as the first exchange for standardised, listed options [MacKenzie & Millo \(2003\)](#)
 - ▶ Before: options were negotiated one by one, over the counter
- The same year: the Black–Scholes and Merton pricing papers [Black & Scholes \(1973\)](#), [Merton \(1973\)](#)
- Traders soon used the formula on the floor: theory changed the market it described [MacKenzie & Millo \(2003\)](#)
- Textbooks: [Franke, Härdle & Hafner \(2019\)](#), [Härdle, Chen & Overbeck \(2017\)](#), [Gatheral \(2006\)](#)

The Formula and Its Authors



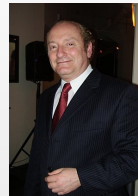
Fischer Black (1938–1995)

Photo: Dalmatine at English Wikipedia;
public domain; Wikimedia Commons



Myron Scholes (2008)

Photo: Nobellaureatesphotographer (2008);
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Robert C. Merton (2006)

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- Key idea: a portfolio of the stock and cash can **replicate** the option, so its price follows from no-arbitrage [Black & Scholes \(1973\)](#)
- Scholes and Merton received the [1997 Nobel Memorial Prize in Economic Sciences](#); Black had died in 1995

Data Used in This Chapter

Series	What it measures	Period	Owner
S&P 500	US equity index, daily close	1990–2026	S&P Dow Jones Indices
SPY	ETF on the S&P 500: open and close	1993–2026	State Street
VIX	30-day implied volatility of the S&P 500	1990–2026	Cboe
VIX9D, VIX3M	the same over 9 days and 3 months	2011 / 2007–2026	Cboe
VVIX	implied volatility of the VIX	2006–2026	Cboe
Bitcoin options	all listed contracts, one dated snapshot	22 Sep 2026	Deribit
DVOL	30-day implied volatility of Bitcoin	2021–2026	Deribit

- Daily data up to 18 September 2026; the option snapshot was taken on 22 September 2026, 07:51 UTC
- Implied volatility is quoted in annualised %; “vol points” = percentage points of volatility

Notation and Prerequisites

- European call and put pay off $(S_T - K)^+$ and $(K - S_T)^+$ at expiry T ; $\tau = T - t$ in years; American options can be exercised at any time up to T
 - ▶ Both pay-offs are convex in S_T : the value of this convexity is what the smile prices
- Forward $F = Se^{(r-q)\tau}$, log-moneyness $k = \ln(K/F)$; ATM: $k \approx 0$; OTM: puts with $K < F$ and calls with $K > F$; ITM: the opposite
 - ▶ Most of this chapter works in k and in total implied variance $w = \sigma_{imp}^2 \tau$
- Model-free bounds (no dividends): $\max(S - Ke^{-r\tau}, 0) \leq C \leq S$ and $\max(Ke^{-r\tau} - S, 0) \leq P \leq Ke^{-r\tau}$
 - ▶ With $r > 0$ an American call on a stock without dividends is never exercised early: it is worth at least $S - Ke^{-r\tau} > S - K$
- Assumed: Brownian motion, Itô's lemma and martingales ([Chapter 11](#)); regression with HAC standard errors ([Chapter 8](#))

Put–Call Parity

- Two portfolios with the same pay-off at T (European options, same K and T , no dividends):
 - ▶ A: one call + cash Ke^{-rT} (worth K at T) $\Rightarrow \max(S_T, K)$
 - ▶ B: one put + one share $\Rightarrow \max(K, S_T)$
- Same pay-off in every state, so the same price today, otherwise there is an arbitrage:
 - ▶ $C_t - P_t = S_t - Ke^{-rT}$ (with dividend yield q : S_te^{-qT} instead of S_t)
- No model for S is needed: parity uses only no-arbitrage [Merton \(1973\)](#)
- Consequence: a call and a put with the same K and T have the **same** implied volatility

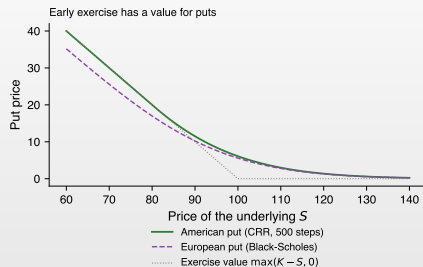
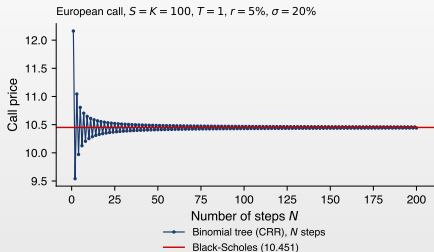
Static Replication and Its Corollaries

- Carr–Madan spanning: any twice-differentiable pay-off $f(S_T)$ and any point F (Taylor with integral remainder; proof in the [Appendix](#)):
 - ▶ $f(S_T) = f(F) + f'(F)(S_T - F) + \int_0^F f''(K)(K - S_T)^+ dK + \int_F^\infty f''(K)(S_T - K)^+ dK$
 - ▶ Priced today (F the forward): $V_0 = e^{-r\tau} f(F) + \int_0^F f''(K) P(K) dK + \int_F^\infty f''(K) C(K) dK$, whatever the model for S [Bakshi, Kapadia & Madan \(2003\)](#)
- **Variance:** $f(S) = -2 \ln(S/F)$ has $f''(K) = 2/K^2$: the strip with weights $1/K^2$ is the variance swap and the VIX (Section 8)
 - ▶ Low strikes get the largest weights, so an expensive put wing makes variance expensive
- **Density:** a butterfly $[C(K - h) - 2C(K) + C(K + h)]/h^2 \rightarrow \partial^2 C / \partial K^2 = e^{-r\tau} q(K)$ [Breedon & Litzenberger \(1978\)](#): the risk-neutral density is a price (Section 7)
- Straddles, spreads and protective puts are finite strips with f'' concentrated at one or two strikes; moments and SVIX: [Appendix](#)

No Arbitrage and the Pricing Measure

- One period, $S \rightarrow uS$ or dS , $d < R = e^{r\Delta t} < u$: $\Delta = \frac{C_u - C_d}{(u - d)S}$ shares plus cash replicate the call, so
 - ▶ $C = \frac{q C_u + (1 - q) C_d}{R}$, $q = \frac{R - d}{u - d}$: q is the unique probability under which S_t/R^t is a martingale
- The real probability of an up move never enters: risk preferences are already in S
- **Fundamental theorem:** no arbitrage $\Leftrightarrow$ an equivalent martingale measure Q exists; complete market $\Leftrightarrow Q$ is unique [Harrison & Pliska \(1981\)](#), [Delbaen & Schachermayer \(1994\)](#)
 - ▶ With jumps or stochastic volatility the market is incomplete: many Q , and option prices carry risk premia (Section 8)
- CRR tree [Cox, Ross & Rubinstein \(1979\)](#): $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$, value backwards from T ; American options: $\max(\text{continuation}, \text{exercise})$ at every node

From the Tree to Black-Scholes



- Left: CRR price of a European call for $N = 1, \dots, 200$ steps; right: American vs European put
- Converges to Black-Scholes (10.451): $N = 50$: 10.411; $N = 200$: 10.441; $N = 1000$: 10.449
 - The error is of order $1/N$ and oscillates: $N = 10$ gives 10.253, $N = 11$ gives 10.611 (strike on or between nodes; Seminar 12, A4)
- American put at $S = K = 100$: 6.090 vs European 5.574; early-exercise premium 0.516: deep in the money it pays to receive K now

The Black–Scholes Model and Equation

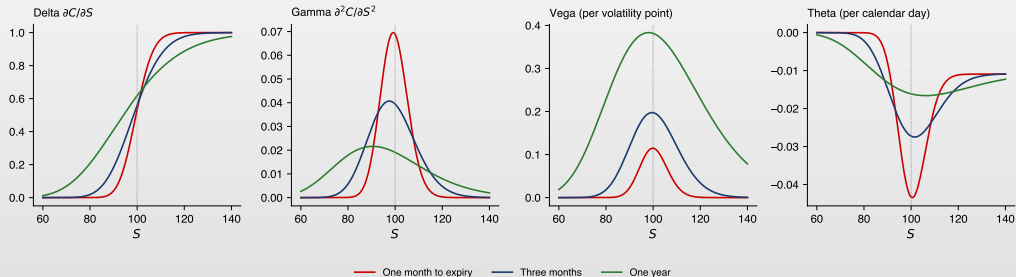
- Geometric Brownian motion $dS_t = \mu S_t dt + \sigma S_t dW_t$: log returns over τ follow the Normal distribution with variance $\sigma^2 \tau$; constant σ and r , continuous trading, no costs, short selling allowed
 - Replicate $V(S, t)$ with a **self-financing** portfolio of Δ_t shares and cash B_t at r : $d\Pi = \Delta dS + rB dt$
- Itô's lemma: $dV = (V_t + \mu S V_S + \frac{1}{2} \sigma^2 S^2 V_{SS}) dt + \sigma S V_S dW_t$
- Choose $\Delta = V_S$, start from $\Pi = V$ ($B = V - S V_S$): $d(V - \Pi) = (V_t + \frac{1}{2} \sigma^2 S^2 V_{SS} - rB) dt$; both dW_t and μ drop out
- A riskless position that costs nothing must earn nothing, hence the **Black–Scholes equation** Black & Scholes (1973), Merton (1973)
 - $V_t + \frac{1}{2} \sigma^2 S^2 V_{SS} + rS V_S - rV = 0$, with $V(S, T) = \text{pay-off}$
- That μ is irrelevant is a result, not an assumption; Feynman–Kac: $V = e^{-r\tau} E^Q[\text{pay-off}]$ with S drifting at r under Q

The Black–Scholes Formula

- European call and put, no dividends (with yield q : replace S by $Se^{-q\tau}$):
 - ▶ $C = S \Phi(d_1) - Ke^{-r\tau} \Phi(d_2), \quad P = Ke^{-r\tau} \Phi(-d_2) - S \Phi(-d_1)$
 - ▶ $d_1 = \frac{\ln(S/K) + (r + \sigma^2/2)\tau}{\sigma\sqrt{\tau}}, \quad d_2 = d_1 - \sigma\sqrt{\tau},$ with Φ the standard Normal distribution function
- Reading the two terms:
 - ▶ $\Phi(d_2)$ = risk-neutral probability that the call ends in the money
 - ▶ $\Phi(d_1)$ = number of shares in the replicating portfolio (the delta)
- Five inputs: S, K, τ, r observed; σ is the only one not observed: it must be estimated or implied

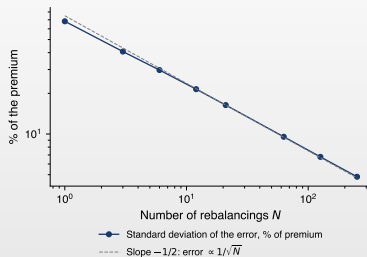
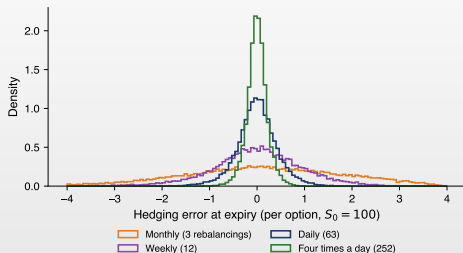
The Greeks: One Identity, All Sensitivities

- Delta $\Delta = \Phi(d_1)$ (put: $\Phi(d_1) - 1$); gamma $\Gamma = \varphi(d_1)/(S\sigma\sqrt{\tau})$, φ the standard Normal density; key identity $S\varphi(d_1) = Ke^{-r\tau}\varphi(d_2)$
- Vega $= S\varphi(d_1)\sqrt{\tau} = \Gamma S^2\sigma\tau$: for one expiry gamma and vega are proportional (Seminar 12, A5–A6); theta $\Theta = rC - rS\Delta - \frac{1}{2}\sigma^2 S^2\Gamma$; rho $= K\tau e^{-r\tau}\Phi(d_2)$



- Call with $K = 100$; short maturities concentrate gamma and theta at the strike, long maturities carry the most vega; vega per vol point, theta per calendar day

Discrete Delta Hedging: Algorithm and Error

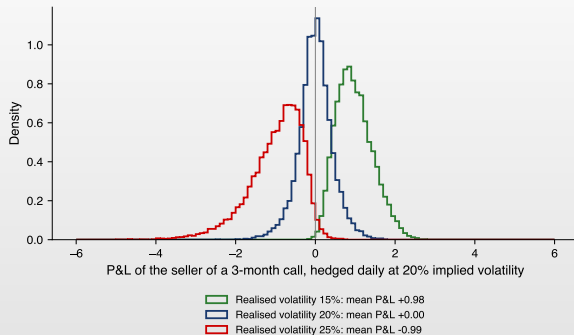


- Sell one call at its Black–Scholes price, buy Δ_0 shares, keep the rest in cash; rebalance N times; **hedging error** = shares + cash – pay-off at expiry (GBM, $\mu = 8\%$, $\sigma = 20\%$, three months, 20 000 paths)
- Mean error near 0; s.d. monthly 40.8%, weekly 21.5%, daily 9.5%, four times a day 4.8% of the premium; worst 1%: -2.68 (weekly), -1.17 (daily)
- Log-log slope -0.48 ; theory (Appendix): s.d. $\approx \sqrt{\pi/4} \mathcal{V}\sigma/\sqrt{N}$, 1.01, 0.44, 0.22 vs simulated 0.96, 0.43, 0.22 ($N = 12, 63, 252$); costs grow like N

Derivation: the Gamma-Weighted Volatility P&L

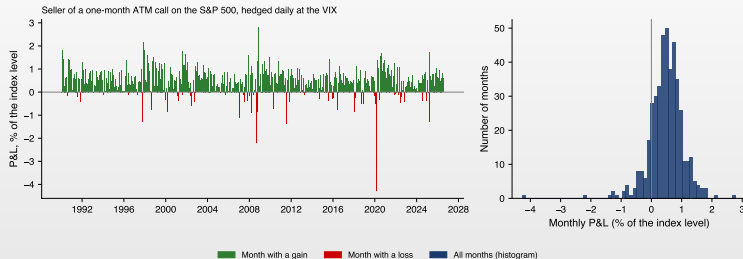
- Sell the option at σ_{imp} and hold $\Delta = \partial_S V^{imp}$; the price moves with σ_{real} : $dS = \mu S dt + \sigma_{real} S dW_t$
- Itô on V^{imp} : $dV = V_t dt + \Delta dS + \frac{1}{2} \Gamma \sigma_{real}^2 S^2 dt$; the equation at σ_{imp} gives $V_t = rV - rS\Delta - \frac{1}{2} \sigma_{imp}^2 S^2 \Gamma$
- Short option + Δ shares + cash at r , self-financing: $d \text{P\&L} = r \text{P\&L} dt + \frac{1}{2} \Gamma_t S_t^2 (\sigma_{imp}^2 - \sigma_{real}^2) dt$, so over the life
 - ▶ $\text{P\&L} = \int_0^T e^{r(T-t)} \frac{1}{2} \Gamma_t S_t^2 (\sigma_{imp}^2 - \sigma_{real,t}^2) dt$: path-dependent, weighted by Γ , large near K and near T
- A short variance swap pays $\sigma_{imp}^2 - \sigma_{real}^2$ with constant weights; a delta-hedged option is a variance bet with random weights
- First-order size: vega $\times$ (implied – realised); here vega = 0.197 per vol point, so 5 vol points $\approx$ 0.99

Selling Volatility at 20%: Three Outcomes



- Three-month call sold at 20% implied volatility, hedged daily; realised volatility 15%, 20% or 25%
- Realised 15%: mean P&L +0.98, positive on 99.8% of paths
- Realised 20%: mean +0.00: only discrete-hedging noise remains (s.d. 0.43)
- Realised 25%: mean -0.99, positive on only 2.2% of paths
- Each mean is close to $\text{vega} \times 5 \text{ vol points}$ (± 0.99): a clean bet on volatility

Thirty-Six Years of Selling S&P 500 Volatility



- Every 21 trading days: sell a one-month ATM call priced with the VIX, hedge it daily at the same volatility; P&L in % of the index; interest and dividends ignored
- 440 months from 1990-01: mean 0.48% a month, Newey–West $t = 16.0$ [Newey & West \(1987\)](#); positive in 86% of months; VIX above the realised volatility of the month in 84% of months (19.3 vs 15.3)
- Worst month -4.25% (2020-03), skewness -1.46 : many small gains, a few large losses; consistent with a **volatility risk premium**, but actual seller returns need option quotes and costs [Bakshi & Kapadia \(2003\)](#)

Implied Volatility

- **Implied volatility** σ_{imp} : the σ that makes the Black–Scholes price equal the market price
 - ▶ $BS(S, K, \tau, r, \sigma_{imp}) = C^{market}$; unique because vega > 0 : the price increases in σ
- Newton's method: $\sigma_{k+1} = \sigma_k - \frac{BS(\sigma_k) - C^{market}}{\mathcal{V}(\sigma_k)}$
 - ▶ Example: $S = K = 100$, three months, $r = 2\%$, market price 4.50; start at $\sigma_0 = 20.00\%$
 - ▶ $BS(\sigma_0) = 4.2322$, $\mathcal{V} = 19.848 \Rightarrow \sigma_1 = 21.3495\%$; the next step changes only the 6th decimal: $\sigma_{imp} = 21.3495\%$
- Traders quote options in implied volatility, not in money: it is comparable across strikes, maturities and assets
- If Black–Scholes were right, all options on one asset would have the **same** implied volatility

After October 1987: the Smile Appears



Amsterdam stock exchange after the crash, 21 October
1987

Photo: [Bart Molendijk / Anefo \(1987\)](#); CC0; [Wikimedia Commons](#)

- Before 1987 index options had a nearly flat implied volatility across strikes [Rubinstein \(1994\)](#)
 - ▶ Since the crash, low-strike puts are much more expensive: the skew
- Option prices showed crash fears from October 1986 to August 1987, but not strongly in the two months immediately before the crash [Bates \(1991\)](#)
- **Smile**: implied volatility as a function of K ; **term structure**: as a function of T ; **surface**: both
- Crash risk on the Bucharest Stock Exchange: [Pele & Mazurencu-Marinescu \(2012\)](#)

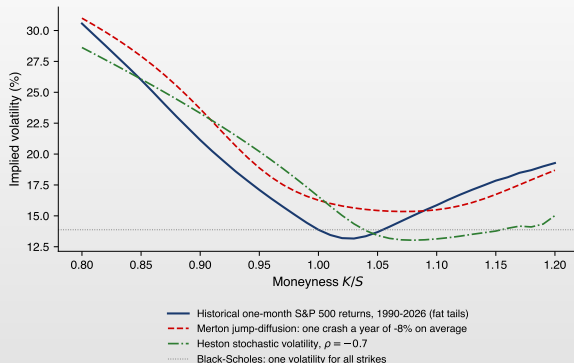
Why Smiles Exist and How Models Produce Them

- **Fat tails:** large moves are more frequent than the Normal distribution allows ([Chapter 1](#)), so deep OTM options are worth more than Black–Scholes says
- **Leverage and volatility feedback:** when prices fall, volatility rises [Christie \(1982\)](#); negative skewness makes low-strike puts dearer
- **Crash fear and demand for insurance:** investors pay a premium for puts; intermediaries need compensation for crash risk they cannot hedge [Bates \(1991\)](#)

Model	Idea	Fits best
Jumps Merton (1976)	Poisson jumps added to the Brownian motion	steep short-dated smiles
Stochastic volatility Heston (1993) , Hull & White (1987)	variance follows its own mean-reverting process, correlated with S	skew and term structure of longer maturities
Local volatility Gatheral (2006)	$\sigma(S, t)$ read off the whole surface (Dupire's formula)	matches every quoted price today, exotic options
Parametric smile, SVI Gatheral & Jacquier (2014)	five parameters per expiry, no dynamics	interpolation and quoting (Section 7)

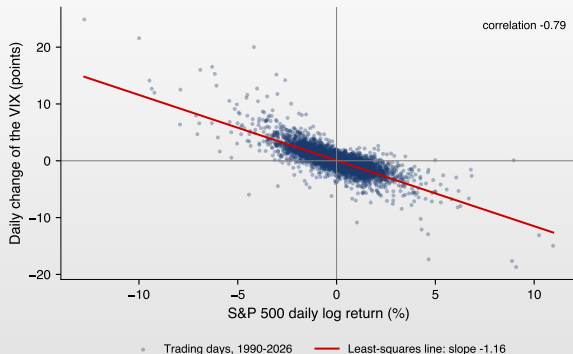
- **Processes:** [Chapter 11](#); the implied-volatility surface in detail: [Franke, Härdle & Hafner \(2019\)](#), [Čížek, Härdle & Weron \(2011\)](#)

Three Sources of a Smile



- One-month implied volatility from the historical S&P 500 monthly returns (rescaled to a martingale, $r = 0$, no risk adjustment), a jump model and a stochastic-volatility model
- History: 9 225 overlapping returns, skewness -1.27 , excess kurtosis 6.8: 30.6% at $K/S = 0.8$, 13.9% ATM, 19.3% at 1.2
- Merton jumps (one crash of about -8% a year): 31.0% vs 16.3%; Heston, $\rho = -0.7$: 28.6% vs 16.6% vs 13.1%
- Risk-neutral index distributions are more left-skewed than realised ones [Jackwerth \(2000\)](#): priced crash risk steepens market smiles further

The Leverage Effect in the Data



- Daily S&P 500 log return vs the change of the VIX on the same day, 1990–2026
- Correlation -0.79 over 9 244 days: implied volatility rises when the market falls
- Slope -1.16 VIX points per 1% return, steeper on down days (-1.39) than on up days (-1.01); mean VIX change $+0.88$ vs -0.76
- The $\rho < 0$ of Heston and the asymmetry of GJR-GARCH ([Chapter 5](#)); low-strike puts gain from both the fall and the rise in volatility, which is why they are expensive

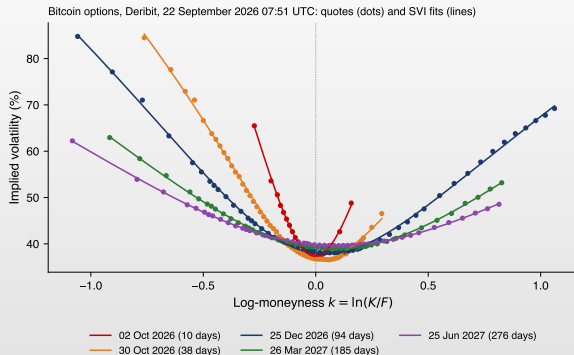
Bitcoin Options and the SVI Smile

- Deribit lists European options on Bitcoin, traded 24/7; **inverse** options: premium and pay-off in BTC; each expiry has its own forward F ; DVOL is its 30-day implied-volatility index [Deribit DVOL](#)
 - ▶ Our snapshot: 22 September 2026, 07:51 UTC; 9 expiries with at least two days left, 375 quoted OTM options; crypto risk: [Pele et al. \(2023\)](#); hedging Bitcoin options: [Matic, Packham & Härdle \(2023\)](#)
- Work with the **total implied variance** $w(k) = \sigma_{imp}^2(k) \tau$ at log-moneyness $k = \ln(K/F)$
 - ▶ SVI (stochastic volatility inspired) [Gatheral & Jacquier \(2014\)](#): $w(k) = a + b \left(\rho(k - m) + \sqrt{(k - m)^2 + s^2} \right)$
 - ▶ a level, b slope of the wings, ρ tilt (negative = put skew), m shift, s curvature at the bottom; wing slopes $b(1 \pm \rho)$
- One SVI fit per expiry by least squares on w ; the fit must then be checked against the no-arbitrage constraints (next slide)

No-Arbitrage Constraints on the Smile

- ▣ **Wings:** Lee (2004) bounds the asymptotic slope, $\limsup_{|k| \rightarrow \infty} w(k)/|k| \leq 2$; for SVI $b(1 + |\rho|) \leq 2$ is necessary, and $w \geq 0$ needs $a + bs\sqrt{1 - \rho^2} \geq 0$
 - ▶ A steeper wing means fatter tails of S_T under Q (moment formula: ▶ Appendix)
- ▣ **Butterfly** (call prices convex in K , density $q \geq 0$), with $w' = \partial_k w$ Gatheral & Jacquier (2014):
 - ▶ $g(k) = \left(1 - \frac{kw'}{2w}\right)^2 - \frac{w'^2}{4} \left(\frac{1}{w} + \frac{1}{4}\right) + \frac{w''}{2} \geq 0$ for all k , plus $d_+(k) = -k/\sqrt{w} + \sqrt{w}/2 \rightarrow -\infty$ as $k \rightarrow \infty$ (Gatheral & Jacquier (2014), Lemma 2.2)
- ▣ **Calendar:** $\partial_\tau w(k, \tau) \geq 0$ for every k : total variance cannot fall with maturity
- ▣ Slice-by-slice least squares imposes none of them; SSVI, the surface version of SVI, has simple sufficient conditions (Gatheral & Jacquier (2014), Theorems 4.1–4.2)

The Bitcoin Smile

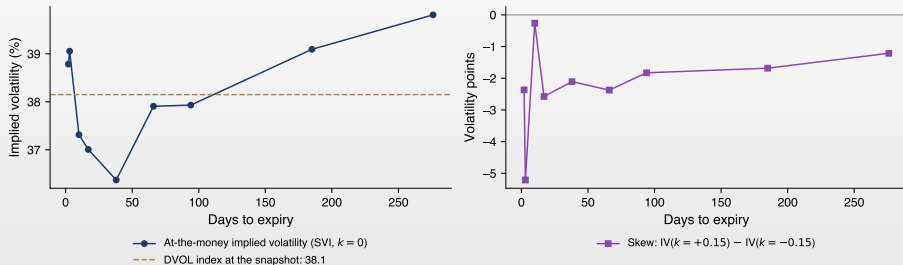


- Mark implied volatilities (mark IV: the reference implied volatility computed by the exchange) of OTM options (dots) and SVI fits (lines) for five expiries
- A true **smile**: both wings rise, unlike the one-sided equity skew; Bitcoin can jump up as well as down [Zulfiqar & Gulzar \(2021\)](#)
- Short expiries have the steepest wings: a jump over a few days matters more relative to $\sigma\sqrt{T}$
- SVI root mean squared error at most 0.76 vol points per expiry; puts slightly dearer than calls: skew from -5.2 to -0.3 vol points

Are the Nine Bitcoin Fits Arbitrage-Free?

- ▣ Wings: largest slopes 0.31 (right) and 0.39 (left), far below 2: Lee's bound holds
- ▣ Butterfly: $\min_k g(k) > 0$ on the grid $k \in [-1.5, 1.5]$ for all 9 expiries (lowest 0.036, 27 November 2026)
- ▣ Calendar: 1 of 8 adjacent pairs crosses: $w(2 \text{ October } 2026) > w(9 \text{ October } 2026)$ on 24% of $k \in [-0.5, 0.5]$, by at most 0.0075
 - ▶ A calendar arbitrage created by the estimator, not quoted by the market: separate fits ignore the cross-expiry restriction
- ▣ Remedy: estimate the surface jointly under the constraints (SSVI, or least squares with $g \geq 0$ and $\partial_\tau w \geq 0$)

Term Structure and Skew

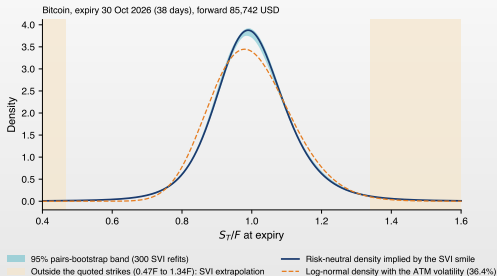


- Left: ATM implied volatility by expiry and the DVOL index; right: skew = implied volatility at $k = +0.15$ minus at $k = -0.15$
- ATM between 36.4% and 39.8%; longest expiry (276 days) 39.8%: upward sloping, calm now and more uncertainty later
- DVOL 38.1 vs ATM 37.0% and 36.4% around 30 days: a variance-strip index need not lie between ATM volatilities
- Skew negative for 9 of 9 expiries, closest to 0 at long maturities; one snapshot is one day

The Risk-Neutral Density: From Prices to a Density

- Breeden & Litzenberger (1978): the second derivative of call prices in the strike gives the risk-neutral density q of S_T
 - ▶ $q(K) = e^{r\tau} \frac{\partial^2 C(K)}{\partial K^2}$; convexity of C in K (no butterfly arbitrage) is exactly $q \geq 0$
- From noisy, discrete quotes this is an ill-posed inverse problem: differentiating twice amplifies the noise; fit a smooth smile (SVI), then differentiate (other estimators: [Appendix](#))
- Uncertainty: pairs bootstrap of the quotes, refit, differentiate: pointwise bands for q and intervals for tail probabilities
- Beyond the last quoted strike the data identify nothing: the tail is an extrapolation; report where the quotes end
- q mixes probabilities and risk aversion: pricing kernel $\propto q/p$ with p the physical density [Aït-Sahalia & Lo \(1998\)](#), [Grith, Härdle & Park \(2013\)](#); its non-monotone shape is the pricing-kernel puzzle [Jackwerth \(2000\)](#)

What the Market Prices for Bitcoin



- Density from the SVI smile of the expiry closest to 30 days (38 days, forward 85 742 USD, ATM 36.4%), 95% pairs-bootstrap band, vs the log-normal density; shaded: outside the 56 quotes ($0.47F$ to $1.34F$)
- Fall of more than 30%: 1.21% [1.08, 1.24] vs 0.14% log-normal, about 8 times more; fall of more than 20%: 4.28% vs 3.27%; rise above 20%: 5.3% vs 5.3%
- Narrow band ($\pm 2.5\%$ at the centre, $\pm 8\%$ at $0.7F$): smoothed mark volatilities, so fitting noise, not bid-ask uncertainty; 0.13 of the 1.21 percentage points lie below the lowest quoted strike
- Prices of insurance, not forecasts: they include the premium for crash risk

 MFM_ch12_btc_rnd

Variance Swaps and the Option Strip

- A **variance swap** pays at T : notional $\times$ (realised variance – strike K_{var}), with realised variance $\frac{252}{n} \sum r_t^2$; the fair strike is the risk-neutral expected variance [Demeterfi, Derman, Kamal & Zou \(1999\)](#)
 - ▶ $K_{var} = \frac{2e^{r\tau}}{\tau} \left(\int_0^F \frac{P(K)}{K^2} dK + \int_F^\infty \frac{C(K)}{K^2} dK \right)$ (spanning with $f = -2 \ln(S/F)$, Section 2)
- Why it works: Itô on $\ln S$ (continuous paths, $r = q = 0$): $-2 \ln(S_T/F) = \int_0^T \sigma_t^2 dt - 2 \int_0^T dS_t/S_t$, and the last term is a Q -martingale [Britten-Jones & Neuberger \(2000\)](#)
 - ▶ “Model-free” means free of a model for σ_t , not of continuity: with crash jumps the strip understates expected realised variance [Carr & Wu \(2009\)](#) (▶ [Appendix](#))
- Buyer of variance = insurance against turbulence; seller = earns a premium, pays in crashes [Carr & Wu \(2009\)](#)

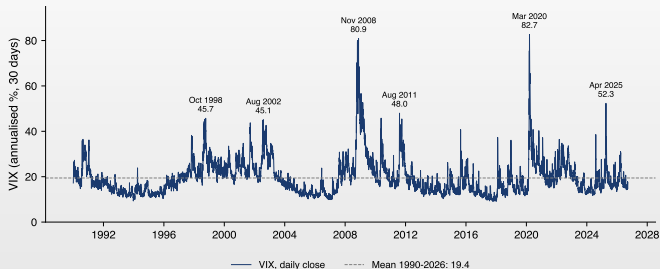
The VIX Index

- Cboe Volatility Index (VIX), since 1993; since 2003 computed from S&P 500 options with the variance-swap formula [Cboe \(2025\)](#)
 - ▶ $\sigma^2 = \frac{2}{T} \sum_i \frac{\Delta K_i}{K_i^2} e^{RT} Q(K_i) - \frac{1}{T} \left(\frac{F}{K_0} - 1 \right)^2$, K_0 = first strike at or below F ; Q = bid-ask mid of the put for $K_i < K_0$, of the call for $K_i > K_0$, the average of both at K_0
 - ▶ $\Delta K_i = (K_{i+1} - K_{i-1})/2$, one-sided at the two ends; options with a zero bid are dropped, and each wing stops after two consecutive zero bids
- Two expiries around 30 days, interpolated to exactly 30 days; $VIX = 100 \sigma$
- VIX futures since 2004, VIX options since 2006 [Cboe \(2025\)](#); the “fear gauge” [Whaley \(2000\)](#)
- Siblings: VIX9D (9 days), VIX3M (3 months), VVIX (the VIX of the VIX)

Worked Example: a VIX for Bitcoin

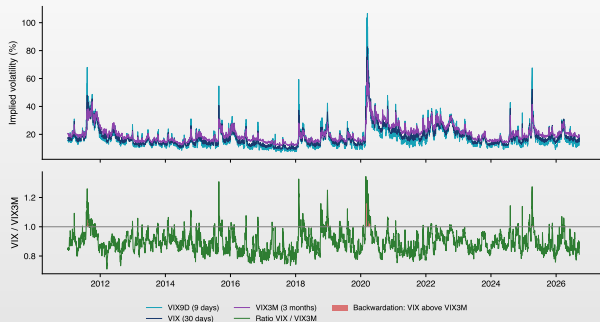
- Cboe rules on our Deribit snapshot: USD mid $Q = \text{BTC mid} \times \text{index } S; F$ and $e^{RT} = F/S$ from Deribit's forward (Cboe: F from parity, R from Treasury yields)
 - ▶ Near expiry: 17 days, 26 OTM options, $\sigma = 39.5\%$; next expiry: 38 days, 56 OTM options, $\sigma = 40.2\%$
- Interpolate the total variances $\sigma_i^2 T_i$ linearly in time to 30 days:
 - ▶ $\sigma_{30}^2 = \frac{1}{30} \left[T_1 \sigma_1^2 \frac{T_2 - 30}{T_2 - T_1} + T_2 \sigma_2^2 \frac{30 - T_1}{T_2 - T_1} \right]$ (days)
- Our Bitcoin VIX: 40.0; Deribit's DVOL at the same moment: 38.1 [Alexander & Imeraj \(2020\)](#); both exceed the ATM volatilities: the variance swap also pays for the wings
- Truncation and strike spacing move the index by at most 0.6 vol points, less than the gap to DVOL (1.9): an index is an estimator, so report its sensitivity (▶ [Appendix](#); Seminar 12, A8)

The VIX, 1990–2026



- Daily close with the six highest peaks at least 500 days apart
- Mean 19.4 (5.6% a month), range 9.1–82.7 (2020-03-16, COVID-19); above 30 on 7.9% of days: sudden spikes, slow decay, as in GARCH ([Chapter 5](#))

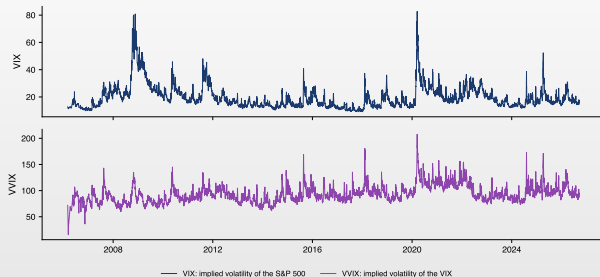
The VIX Term Structure



- Top: implied volatility over 9 days, 30 days and 3 months; bottom: ratio VIX / VIX3M, 2011–2026
- Normally upward sloping (**contango**): mean $VIX / VIX3M = 0.89$; VIX9D below the VIX on 73% of days
- Inverted (**backwardation**) on 7.6% of days, always in crises: fear is concentrated in the near term
- Next 21 days after backwardation: return 2.88%, realised volatility 26.7%, vs 0.80% and 13.6% in contango: high risk, high expected return after panics [Bollerslev, Tauchen & Zhou \(2009\)](#)

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The Volatility of Volatility

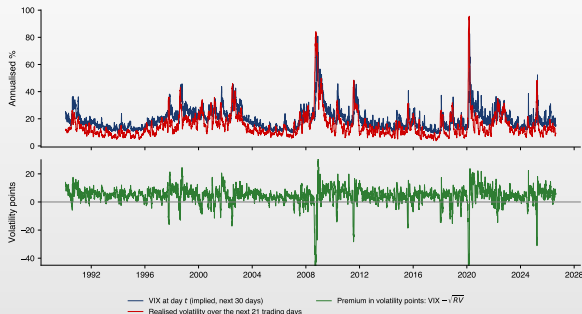


- VIX (top) and VVIX (bottom), daily closes, 2006–2026
- Mean VVIX 93.5: the VIX is expected to move about 5.3 times more than the index, in relative terms
- Highest value 207.6 on 2020-03-16, the day of the VIX peak; daily changes of VIX and VVIX correlate 0.71
- Volatility is itself random: the core of stochastic-volatility models such as Heston (1993)

The Variance Risk Premium

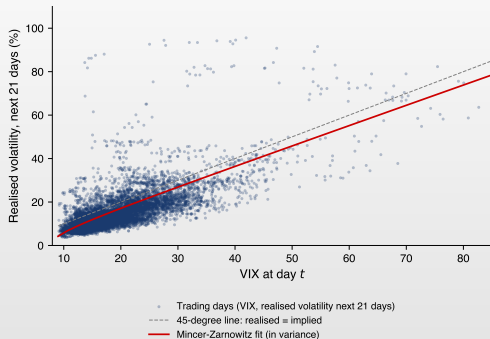
- Variance risk premium (VRP) Carr & Wu (2009), Bollerslev, Tauchen & Zhou (2009); three versions that must not be confused:
 - ▶ **Economic (ex ante):** $E_t^Q[RV_{t,t+21}] - E_t^P[RV_{t,t+21}]$; VIX_t^2 estimates the first term, the second needs a forecast
 - ▶ **Ex post** (used here): $VIX_t^2 - RV_{t,t+21}$, the realised P&L of the seller of a variance swap, known only at $t + 21$
 - ▶ **Ex-ante proxy:** $VIX_t^2 - RV_{t-21,t}$, past variance as the forecast of E_t^P ; known at t , so it can serve as a predictor (Section 11)
 - ▶ $RV_{t,t+21} = \frac{252}{21} \sum_{i=1}^{21} r_{t+i}^2$ (annualised, in $\%^2$)
- Positive on average if investors dislike states with high volatility, which are also bad states for wealth
- Overlapping 21-day windows are autocorrelated: Newey–West with at least 21 lags Newey & West (1987); persistent volatility can extend the dependence, so check longer bandwidths

Implied vs Realised Volatility, S&P 500



- Top: VIX and the volatility realised over the next 21 trading days; bottom: their difference, 1990–2026
- 9 203 days: mean VIX 19.4, mean realised volatility 15.3; premium 4.09 vol points (Newey–West s.e. 0.27); in variance 113 %², $t = 5.5$
- Positive on 86% of days; the exceptions are the crashes: worst -72.0 vol points, window starting 2020-02-20 (COVID-19)
- Same message as the delta-hedged option: sellers of volatility are paid on average and lose in crises

Is the VIX a Good Forecast?



- Mincer–Zarnowitz [Mincer & Zarnowitz \(1969\)](#):
 $RV_{t,t+21} = \alpha + \beta VIX_t^2 + \varepsilon_t$; unbiased means $\alpha = 0$, $\beta = 1$;
 Newey–West errors, 21 lags
- $\hat{\alpha} = -51.9$ (34.4), $\hat{\beta} = 0.86$ (0.10), $R^2 = 0.37$; separately
 neither differs from (0, 1): $t = -1.51$ and -1.34
- But the estimates correlate -0.90 : joint Wald $\chi^2_2 = 41.7$,
 $p = 8.9 \cdot 10^{-10}$: unbiasedness rejected; most points lie
 below the 45-degree line
- Past realised variance alone: $R^2 = 0.29$; with both, the VIX
 keeps its weight (0.77), past variance adds little (0.08, s.e.
 0.13)

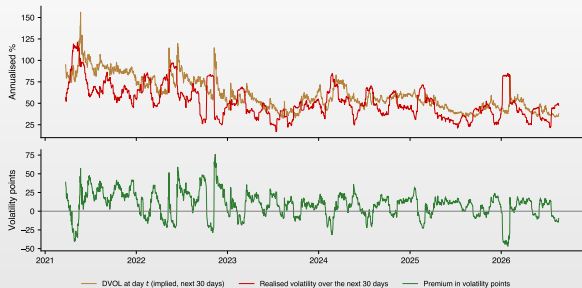
Logs, Instruments and HAR-RV: Inference Done Right

Specification	$\hat{\beta}$	s.e.	$t(\beta = 1)$	Note
Levels, OLS	0.86	0.10	-1.34	joint Wald $p = 8.9 \cdot 10^{-10}$
Logs, OLS	1.06	0.035	1.65	intercept -0.94 , $R^2 = 0.57$
Logs, instrumental variables	1.02	0.042	0.60	first-stage $t = 37.7$

- Levels are dominated by 2008 and 2020; in logs the slope is close to 1 and the premium is a proportional shift (intercept < 0)
- VIX^2 measures expected variance with error, which attenuates OLS: instrument it with VIX^2 21 days earlier [Christensen & Prabhala \(1998\)](#); the slope barely moves (valid only if the lag is uncorrelated with the forecast error, untestable with one instrument)
- Against HAR-RV [Corsi \(2009\)](#): in sample the HAR terms add information ($\chi^2_3 = 14.2$, $p = 0.003$); out of sample from 2000 the VIX is about 6% better in QLIKE loss, Diebold–Mariano $p = 0.35$ [Diebold & Mariano \(1995\)](#): not significant

► Test details: [► Appendix](#)

The Variance Risk Premium of Bitcoin



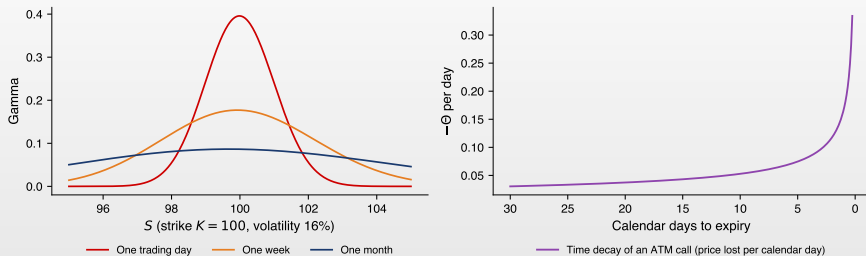
- DVOL and the Bitcoin volatility realised over the next 30 calendar days, 2021–2026; bottom: their difference
- 1 975 days: mean DVOL 60.3 vs realised 51.7; premium 8.5 vol points (s.e. 1.6, $t = 5.3$): twice the S&P 500 premium in vol points, similar relative to the level
- Positive on 71% of days, less often than for the S&P 500; worst -46.9 vol points (window starting 2026-01-28)
- A short sample: 66 non-overlapping 30-day periods; a Bitcoin VIX and its premium: Alexander & Imeraj (2020)

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Options That Live One Day

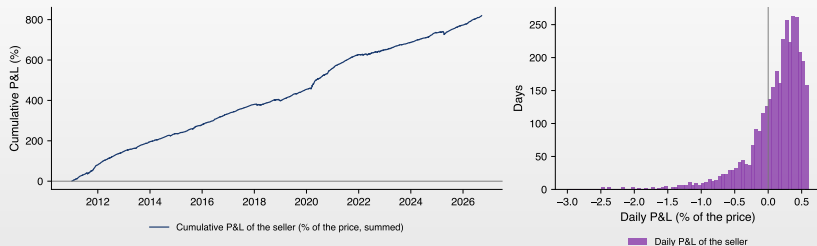
- **0DTE** (zero days to expiry) options: traded on the day they expire [Cboe 0DTE](#)
 - ▶ S&P 500 options now expire every trading day, so a 0DTE contract exists every day
- More than 75% of retail trades in S&P 500 options involve 0DTE contracts, and retail investors lose money on them [Beckmeyer, Branger & Gayda \(2023\)](#)
 - ▶ 0DTE options were 48% of index-option trading in August 2023 [Brogaard, Han & Won \(2023\)](#)
- More 0DTE trading is followed by higher index volatility, mostly through speculative retail trades [Brogaard, Han & Won \(2023\)](#); ultra-short pricing needs skewness and kurtosis corrections [Bandi, Fusari & Renò \(2023\)](#)
- Theory from Section 4: with $\tau \rightarrow 0$ gamma and theta explode at the money

Gamma and Time Decay Near Expiry



- Left: gamma of a call with $K = 100$ one day, one week and one month before expiry; right: daily time decay of an ATM call
- ATM gamma one day before expiry 0.396 vs 0.086 one month before: 4.6 times larger ($= \sqrt{21}$); a 1% move in the last day changes delta by about 0.4
- Time decay 0.167 in the last day vs 0.030 with 30 days left, about 5.5 times more: the seller collects fast decay and carries large gamma risk

Selling a Same-Day Straddle: a Stylised Test



- SPY, 2011–2026: sell an ATM straddle at the open, settle at the close; premium = $\sqrt{2/\pi} \sigma_{day} S$, σ_{day} from the previous VIX9D scaled to the session (share 0.74 of daily variance, estimated on 1993–2010)
- 3950 days: mean premium 0.76% vs mean absolute move 0.55%; the seller wins on 75% of days, 20.7 basis points a day
- Worst day –8.25% (2025-04-09), about 11 premia; stylised pricing measures the premium inside VIX9D, not traded 0DTE prices [Bandi, Fusari & Renò \(2023\)](#), [Beckmeyer, Branger & Gayda \(2023\)](#)

Case Study: the Research Question

- **Question Back, Crotty & Kazempour (2022):** option prices give lower bounds on the equity premium; are they valid and tight conditionally, and do they forecast returns out of sample?
 - ▶ Why: a tight bound is a real-time equity-premium forecast with no estimated parameter; **Martin (2017)** could not reject $\alpha = 0$, $\beta = 1$ on 1996–2012, a test with low power
- **Martin bound** (eq. 3), $R_{t,T}$ the gross market return, $R_{f,t,T}$ the gross riskless return, var_t^* the risk-neutral variance ($\text{SVIX}_{t,T}^2$ not annualised here):
 - ▶ $E_t[R_{t,T}] - R_{f,t,T} \geq R_{f,t,T} \text{SVIX}_{t,T}^2$, $\text{SVIX}_{t,T}^2 = \text{var}_t^*(R_{t,T}/R_{f,t,T})$
 - ▶ Valid if $\text{cov}_t(M_{t,T}R_{t,T}, R_{t,T}) \leq 0$ ($M_{t,T}$ the stochastic discount factor); tight if this covariance is zero, e.g. a log-utility investor who holds the market
- **Chabi-Yo–Loudis bound** **Chabi-Yo & Loudis (2020)**: built from the first four risk-neutral moments, free of preference parameters in its restricted form; almost always above the Martin bound (Figure 3.1)
- **Link to this chapter:** SVIX is the static-spanning strip with $f = S^2$ (Section 2 and [Appendix](#))
 - ▶ Realised slackness $s_{t,T} = R_{t,T}^e - b_{t,T}$ (excess return minus bound): valid means $E_t[s_{t,T}] \geq 0$, tight means $E_t[s_{t,T}] = 0$

Case Study: Data and Method

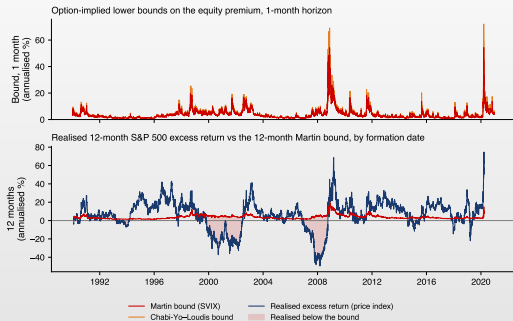
- **Data** (Section 3, [Appendix C](#)): daily Martin and Chabi-Yo–Loudis bounds, January 1990 – December 2020, from S&P 500 index options, interpolated to 30, 90, 180 and 360 days; S&P 500 excess returns over 21, 63, 126 and 252 trading days, annualised, in % (filters: [Appendix](#))
- **Moments** (eq. 10): for any nonnegative z_t known at t , $E[(R_{t,T}^e - b_{t,T}) z_t] \geq 0$; z_t = a constant and nine Welch–Goyal predictors made positive [Welch & Goyal \(2008\)](#) (Table 3)
- **Kodde–Palm tests** [Kodde & Palm \(1986\)](#) (eqs. 11–14): $\bar{\lambda}$ the 10 sample means, Σ their Hansen–Hodrick covariance [Hansen & Hodrick \(1980\)](#), with lags equal to the days in the horizon
 - ▶ $D_1 = \min_{\lambda \geq 0} (\lambda - \bar{\lambda})^\top \Sigma^{-1} (\lambda - \bar{\lambda})$ tests validity ($H_0: \lambda_0 \geq 0$); $D_2 = \bar{\lambda}^\top \Sigma^{-1} \bar{\lambda} - D_1$ tests tightness ($H_0: \lambda_0 = 0$ vs $\lambda_0 \geq 0$); both follow mixtures of χ^2 laws
- **Forecasting**: bias-corrected predictive regressions [Amihud & Hurvich \(2004\)](#) (Tables 5, 6) and monthly out-of-sample forecasts against the historical mean (Table 8; R_{OS}^2 : out-of-sample R^2 against the historical mean; protocol: [Appendix](#))

Case Study: Results (Tables 1, 3, 5, 6 and 8)

Horizon	Mean R^e (%)	Mean b (%)	D_1	D_2	$\hat{\beta}$ (s.e.)	R_{OS}^2 (%)
1 month	8.76	3.81	0.00	29.93	1.36 (1.23)	-1.2
3 months	8.73	3.86	0.00	39.51	1.39 (1.47)	-1.2
6 months	8.67	3.83	0.00	49.28	1.78 (0.85)	-0.2
12 months	8.86	3.68	0.00	71.21	1.04 (0.88)	-3.0

- Source: [Back, Crotty & Kazempour \(2022\)](#), *Journal of Financial Economics*, Tables 1, 3, 5, 6 and 8
- **Validity not rejected:** all 10 sample moments are positive, so $D_1 = 0$ and $p = 100\%$ at every horizon; this does not prove validity in every state
- **Not tight:** D_2 exceeds 27.13, the upper 0.1% Kodde–Palm critical value, at every horizon (finite-sample p : 3.6, 0.4, 2.6, 8.3%); mean slackness 4.8 to 5.2 points a year: the bound is less than half of the realised premium
- **Forecasting:** slopes positive but imprecise (Table 5); with the nine predictors as controls a one-s.d. rise in the 12-month bound goes with a 6.6-point higher 12-month excess return, but bootstrap p -values are 20.7, 0.0, 0.2, 9.5% (Table 6)
- **Out of sample:** the Martin bound loses to the historical mean at every horizon; the best case, Chabi-Yo–Loudis at 6 months, reaches 4.3%, not significant (Table 8): low power is one possible explanation

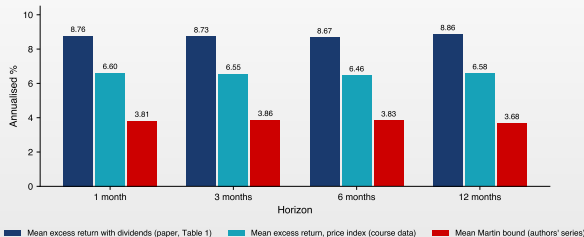
Case Study: the Bound and the Realised Premium



- Top: the authors' daily 1-month bounds, 1990–2020 ([replication package](#)); bottom, course data: 12-month excess return of the S&P 500 price index vs the 12-month Martin bound
- 1-month bound: median 2.7%, peak 54.4% (16 March 2020); 12-month peak 21.1% (20 November 2008): a risk-neutral variance, it jumps with the VIX in crises
- Chabi-Yo-Loudis at or above Martin on 99.99% of days: negative skewness and fat tails raise it
- Realised 12-month excess return below the bound on 33% of dates (2000–2002, 2007–2008): the bound restricts $E_t[R_{t,T}^e]$, not each realisation
- Correlation bound–realised 0.15 (overlapping, descriptive): return noise dwarfs the bound, hence imprecise slopes and negative R_{OS}^2

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Case Study: Mean Premium and Mean Bound



- Dark blue: Table 1 of the paper (S&P 500 with dividends); teal: course data (price index, the same formation days: 7789 at 1 month, 7598 at 12 months); red: mean of the authors' Martin bound
- Mean slackness on the course data 2.79, 2.69, 2.63, 2.91 points a year vs 4.8–5.2 in the paper: the gap of 2.2–2.3 points is mostly the missing dividend yield
- The mean bound covers 41–44% of the premium with dividends, 56–59% of the price-index premium; single realisations fall below it on 41% (1 month) to 33% (12 months) of days, the average never (Seminar 12, C3)

Case Study: Replication, Limitations and an Open Extension

- **Replicate** (Seminar 12, C3): the authors' daily bound series ([replication package](#)), S&P 500 daily closes of the course data, the daily risk-free rate ([Kenneth French Data Library](#)), the Welch–Goyal predictors ([Amit Goyal's data page](#))
 - ▶ Rebuild Table 1 and D_1 , D_2 of Table 3; the course index has no dividends, so excess returns and slackness are lower: a data limitation
- **Limitations stated by the authors**
 - ▶ A short series, limited to the years after 1990 by the need for option prices: the out-of-sample tests have little power (Section 6)
 - ▶ Validity holds even when the bound is replaced by its time-series mean, so it says nothing about predictive power (Section 4.3)
 - ▶ More recent bounds and formulas for risk premia are not tested (Introduction)
- **Open extension**: is the Martin bound still valid on 2021–2026 (the 2022 bear market, the 0DTE era of Section 9)? The course data have VIX, not SVIX: VIX weights OTM options by $1/K^2$ and exceeds SVIX when the left tail is priced ([Martin \(2017\)](#), Result 3); is a VIX-based bound still valid, and can the Kodde–Palm test detect it if not?

An Open Question: Does the VRP Still Predict Returns?

- [Bollerslev, Tauchen & Zhou \(2009\)](#), Table 2: monthly data 1990–2007, $y_t^{(h)} = \frac{12}{h} \sum_{j=1}^h (r_{m,t+j} - r_{f,t+j})$ on $\text{VRP}_t = \text{VIX}_t^2/12 - \text{RV}_t$
 - ▶ RV_t : realised variance of month t ; at $h = 3$: slope 0.47, Hodrick $t = 2.86$, adjusted $R^2 = 6.82\%$, the peak of a hump-shaped horizon profile
- [Goyal, Welch & Zafirov \(2024\)](#): 29 equity-premium predictors proposed after 2008, re-examined with data to 2021: over one third lose in-sample significance, half of the rest fail out of sample
- Open: is VRP predictability a stable price of variance and jump risk, or a property of 1990–2007? The 2008–2026 sample adds two crashes and the 0DTE era
- Formally: $H_0 : \beta_3 = 0$ vs $H_1 : \beta_3 > 0$ in $y_t^{(3)} = \alpha + \beta_3 \text{VRP}_t + u_t$
 - ▶ Out of sample: H_0 : the VRP forecast does not lower the mean squared prediction error of the historical mean; Clark–West statistic [Clark & West \(2007\)](#)

The Discovery Loop and What the Researcher Must Check

- **Literature:** ask an assistant for peer-reviewed tests of VRP predictability since 2009 (DOI, sample, table)
 - ▶ Check: resolve every DOI on Crossref and compare the cited table with the paper, digit by digit
- **Hypotheses:** time-varying risk aversion, jump-risk compensation, or a small-sample artefact of persistent regressors and overlapping returns
 - ▶ Check: each must imply something testable for the horizon profile, for 2008–2026 and for non-US markets
- **Data and code:** build $VIX^2/12$ and monthly RV from the daily VIX and S&P 500 closes of the course data, reproduce Table 2 ($h = 3$) before extending anything
 - ▶ Check timing (VIX at the end of month t , returns from $t + 1$) and match the published means (implied 33.23, realised 14.93)
- **Robustness and critique:** a dated pre-analysis plan (horizons, start date, benchmark, Holm correction); a hostile-referee prompt
 - ▶ Check overlap (Hodrick errors [Hodrick \(1992\)](#)), horizon and start-date mining (R_{OS}^2 against the start date), the weight of 2008 and 2020; full prompts: ▶ [Appendix](#)

Mini-Case: Replicate, Then Extend

- A typical assistant proposal: "the VRP predicts the quarterly equity premium; extend the 2009 result to today". We test it on the course data
- Replication, February 1990 to September 2007, $h = 3$, daily data: $\hat{\beta}_3 = 0.42$, $t_{\text{Hodrick}} = 2.25$, adjusted $R^2 = 4.6\%$ (published: 0.47, 2.86, 6.82%)
 - ▶ Mean implied variance 33.1 is close to 33.23; mean realised 20.6 vs 14.93: daily close-to-close data, a stated limitation
- Extension to May 2026 (436 months): $\hat{\beta}_3 = 0.09$, $t_{\text{Hodrick}} = 1.33$, adjusted $R^2 = 1.0\%$; all horizons with four kinds of standard errors: [▶ Appendix](#)
- Out of sample from January 2008, 221 expanding-window forecasts: $R_{OS}^2 = 1 - \sum_t (y_t - \hat{y}_t)^2 / \sum_t (y_t - \bar{y}_t)^2 = -22.2\%$ [Campbell & Thompson \(2008\)](#); Clark–West $t = 0.64$ ($p = 0.26$): H_0 not rejected
- Reading: sign and size replicate in-sample; the longer sample does not support stable predictability. Outliers, measurement or a vanished premium: that is the open part

Your Turn: a Project Seed

- Question: does the VRP predict the equity premium after 2008, and does the Bitcoin premium of Section 8 (DVOL) predict Bitcoin returns?
- Replicate first: [Bollerslev, Tauchen & Zhou \(2009\)](#), Table 2, $h = 3$: slope 0.47, $t = 2.86$, adjusted $R^2 = 6.82\%$; explain every gap
- Deliverables
 - ▶ A dated pre-analysis plan; the replication table; out-of-sample results for all horizons with a Holm correction
 - ▶ Robustness (5-minute variance, subsamples, without 2008 and 2020); a referee-style critique and your reply
- Log each prompt that shaped a decision in `AI_USE.md`; each invented reference, wrong number or look-ahead caught in `AI_ERRORS.md`, with the test that caught it ([templates](#))

Key Takeaways

- Parity, bounds and static spanning need no model; the tree and the Black–Scholes equation price by dynamic replication
- The Greeks measure exposures; discrete hedging leaves an error of order $1/\sqrt{N}$
- A delta-hedged option is a bet on realised vs implied variance; implied variance exceeded realised variance on average on the S&P 500 and on Bitcoin
- The surface and the risk-neutral density are estimates: impose no-arbitrage constraints and report bands
- The VIX is a strip estimator with jump and truncation biases; as a forecast it needs joint tests, logs and HAC loss comparisons
- Option-implied bounds on the equity premium are valid but not tight in 1990–2020: evidence rests on average slackness, not on forecasting power

Frontier, Next Chapter and Further Study

▣ Frontier

- ▶ Andersen, Fusari & Todorov (2015): a left-tail factor in index-option surfaces, not spanned by volatility, predicts future equity and variance premia
- ▶ Bali et al. (2023): nonlinear machine learning on option and stock characteristics predicts equity-option returns out of sample
- ▶ Open: jointly estimated, arbitrage-free surfaces with valid confidence bands; the pricing of 0DTE options

▣ Next: Chapter 13, machine learning for financial prediction, with volatility and option-implied features among the inputs

▣ Further reading: Franke, Härdle & Hafner (2019), Härdle, Chen & Overbeck (2017), Čížek, Härdle & Weron (2011), Gatheral (2006); pricing kernels: Ait-Sahalia & Lo (1998), Grith, Härdle & Park (2013); crypto options: Matic, Packham & Härdle (2023), Alexander & Imeraj (2020), Pele et al. (2023)

- ▶ Quantinar:  XFG Advanced Methods in Quantitative Finance,  Pricing Kernels and Risk Premia,  Hedging Cryptocurrency Options
- ▶ Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_12

Appendix: Proof of Spanning, Moments and SVIX

- Proof for $S_T > F$: $\int_F^{S_T} f''(K)(S_T - K) dK = f(S_T) - f(F) - f'(F)(S_T - F)$ by parts; the put integral vanishes; symmetric for $S_T < F$
- Take $e^{-r\tau} E^Q[\cdot]$ with F the forward (a forward contract costs 0): a bond, a forward and a strip of OTM options with weights $f''(K)$ replicate any smooth pay-off
- **Moments:** $f = [\ln(S/F)]^n$, $n = 2, 3, 4$, price the raw risk-neutral moments of the log return; centring and standardising gives volatility, skewness and kurtosis [Bakshi, Kapadia & Madan \(2003\)](#)
- **SVIX** [Martin \(2017\)](#), from $f = S^2$: $SVIX_t^2 = \frac{2}{\tau R_f S_t^2} \left[\int_0^F P dK + \int_F^\infty C dK \right] = \text{var}_t^Q(R_T/R_f)/\tau$, with R_T, R_f gross returns
 - If $\text{cov}_t(M_T R_T, R_T) \leq 0$ (M_T the stochastic discount factor): $\frac{1}{\tau}(E_t R_T - R_f) \geq R_f SVIX_t^2$, a lower bound on the equity premium (case study, Section 10)

Appendix: Why $1/\sqrt{N}$? The Discrete Hedging Error

- Over one interval $\Delta t = T/N$ the hedged book earns $\frac{1}{2}\Gamma S^2 [\sigma^2 \Delta t - (\Delta S/S)^2]$: time decay against realised squared return
- Mean 0 under the model to leading order in Δt ; $\text{Var}[(\Delta S/S)^2] \approx 2\sigma^4 \Delta t^2$ and the terms are approximately uncorrelated, so $\text{Var}(\text{error}) \approx \frac{1}{2}\sigma^4 \Delta t E \int_0^T \Gamma_t^2 S_t^4 dt$
- ATM call: $E[\Gamma_t^2 S_t^4] \approx \frac{K^2}{2\pi\sigma^2\sqrt{T^2-t^2}}$ integrates to $K^2/(4\sigma^2)$, so s.d. $\approx \sigma K \sqrt{T/(8N)} = \sqrt{\pi/4} \mathcal{V}\sigma/\sqrt{N}$
 - With vega $\mathcal{V} = 19.72$ per unit: theory 1.01, 0.44, 0.22 vs simulated 0.96, 0.43, 0.22 for $N = 12, 63, 252$
- Asymptotic theory of the replication error: [Bertsimas, Kogan & Lo \(2000\)](#); it fails with jumps (the error does not vanish as $N \rightarrow \infty$), with stochastic volatility (vega risk) and with costs

Appendix: the Moment Formula for the Wings

- Moment formula of [Lee \(2004\)](#): $\beta_R = \limsup_{k \rightarrow \infty} w(k)/k \in [0, 2]$ and $\tilde{\rho} = \frac{1}{2\beta_R} + \frac{\beta_R}{8} - \frac{1}{2}$
 - ▶ $\tilde{\rho} = \sup\{p : E^Q[S_T^{1+p}] < \infty\}$: a steeper wing means fatter tails; the left wing mirrors it with negative moments
- Bitcoin: the largest right slope 0.31 gives $\tilde{\rho} = 1.18$; a property of SVI's linear extrapolation beyond the last strike, not of the data
- For SVI a right-wing slope below 2 also gives the tail condition $d_+(k) \rightarrow -\infty$ of [Gatheral & Jacquier \(2014\)](#), Lemma 2.2

Appendix: Estimators of the Risk-Neutral Density

- Fit a smile (SVI) and differentiate the implied call prices twice, as in the lecture
- Kernel regression of C on (K, τ) [Aït-Sahalia & Lo \(1998\)](#); the same under monotonicity and convexity in K [Aït-Sahalia & Duarte \(2003\)](#)
- Smoothness-penalised implied trees [Jackwerth & Rubinstein \(1996\)](#); shape-invariant pricing kernels [Grith, Härdle & Park \(2013\)](#)
- Tails beyond the quotes: SVI wings or an extreme-value tail; either way an assumption, not an estimate

Appendix: What the Strip Measures When Prices Jump

- With jumps J : the strip prices $\int \sigma_t^2 dt + 2 \sum (e^J - 1 - J)$, but realised variance is $\int \sigma_t^2 dt + \sum J^2$
 - ▶ Gap $2E^Q \sum (e^J - 1 - J - J^2/2) \approx E^Q \sum J^3/3$: negative for crashes, so the strip understates expected realised variance Carr & Wu (2009)
- Merton model of Section 6 (one jump a year, mean -8%): strip 17.82 vs $\sqrt{E^Q[\text{QV}]} = 18.00$ vol points, -2.0% in variance

Appendix: Truncation and Discretisation of the Bitcoin Index

- Only a finite set of strikes is quoted: truncating the strip biases the index down, a coarse grid biases it up [Jiang & Tian \(2005\)](#)
- Bitcoin, same two expiries, 30-day index: quoted strikes 40.03; SVI prices at the same strikes 39.99
 - ▶ SVI on a dense grid inside the quoted range 39.78; extended to $k \in [-4, 4]$ with the SVI wings 40.34
- The near expiry is quoted only from $0.76F$ to $1.17F$: its volatility rises from 39.05 to 40.00 when the wings are added
- The rest of the gap to DVOL reflects its quote selection and methodology, not quantified here

Appendix: Encompassing and Out-of-Sample Tests

- ▣ Heterogeneous autoregressive model (HAR-RV) [Corsi \(2009\)](#): past daily, weekly and monthly realised variance; encompassing regression in logs [Busch, Christensen & Nielsen \(2011\)](#)
 - ▶ VIX slope 0.90 (0.056); HAR terms jointly $\chi^2_3 = 14.2$, $p = 0.003$; R^2 : VIX 0.570, HAR 0.474, both 0.575
- ▣ Out of sample from 2000 (6 697 days; coefficients re-estimated every 21 days on targets already observed)
 - ▶ QLIKE loss $L = RV/\hat{\sigma}^2 - \ln(RV/\hat{\sigma}^2) - 1$, robust to noise in the RV proxy [Patton \(2011\)](#)
 - ▶ Loss ratio VIX² / HAR: 0.94, Diebold–Mariano $t = -0.93$ ($p = 0.35$) [Diebold & Mariano \(1995\)](#); MZ-corrected VIX: 0.95 ($p = 0.35$)
- ▣ Targets overlap by 20 of 21 days: the Diebold–Mariano variance is HAC with 21 lags; instrumental variables by just-identified GMM with a HAC covariance

Appendix: Does the Ex-Ante VRP Predict Returns? Overlapping-Horizon Inference

Horizon	$\hat{b}$	t_{NW}	t_{HH}	$t_{Hodrick}$	p_{boot}	R^2 (%)
$h = 1$	0.07	0.33	0.36	0.41	0.35	0.2
$h = 3$	0.09	0.69	0.67	1.33	0.07	1.2
$h = 6$	0.03	0.49	0.48	0.65	0.26	0.3
$h = 12$	-0.02	-0.64	-0.79	-0.54	0.69	0.2

- Monthly 1990–2026: $\frac{12}{h} \sum_{j=1}^h r_{t+j}^e = a + b \text{VRP}_t + u_t$, $\text{VRP}_t = \text{VIX}_t^2/12 - \text{RV}_t$, ex-ante proxy with past RV_t as the forecast [Bollerslev, Tauchen & Zhou \(2009\)](#)
- If one-month errors are serially uncorrelated, overlap makes u_t an $\text{MA}(h-1)$: Newey–West (h lags), Hansen–Hodrick [Hansen & Hodrick \(1980\)](#), Hodrick 1B [Hodrick \(1992\)](#) (better size); these bandwidths are choices, so check their sensitivity
- p_{boot} : one-sided, VRP simulated as $\text{AR}(1)$ ($\hat{\phi} = 0.13$) under $b = 0$, residuals resampled jointly [Stambaugh \(1999\)](#); no horizon is significant on the full sample (Section 11 revisits 1990–2007)

Appendix: Case Study, Filters and Forecasting Protocol

- ▣ Option data: Cboe for 1990–1995 and 2020–2021, OptionMetrics for 1996–2019
- ▣ Filters: standard options only (no weekly or quarterly), bid > 0, the OTM option at each strike, 7 to 550 days, at least 10 strikes per expiry
- ▣ Returns: S&P 500 with dividends minus the Fama–French risk-free rate, up to March 2021
- ▣ Out of sample (Table 8): monthly forecasts against the expanding post-1926 market mean, 60-month initial window
 - ▶ $R_{OS}^2 = 1 - \sum \varepsilon_t^2 / \sum \nu_t^2$, ε_t and ν_t the errors of the forecast and of the benchmark; Diebold–Mariano tests Diebold & Mariano (1995)

Appendix: The Discovery Loop, Reusable Prompts (1/2)

- **Literature:** "List peer-reviewed tests of VRP return predictability since 2009: DOI, sample, horizon, table, headline estimate."
 - ▶ Resolve every DOI on Crossref and read the cited table yourself; search with Semantic Scholar, Google Scholar, Elicit
- **Hypotheses:** "Give three competing explanations of VRP predictability and the testable implication of each for the horizon profile, for 2008-2026 and for non-US markets."
- **Data and code:** "Write Python that builds VIX squared/12 and monthly realised variance from the daily VIX and S&P 500 closes of the course data and reproduces Table 2 of BTZ (2009), $h = 3$, with Hodrick (1992) standard errors."
 - ▶ Any assistant (Claude, ChatGPT, Gemini, GitHub Copilot); replicate the published number before extending anything

Appendix: The Discovery Loop, Reusable Prompts (2/2)

- **Robustness**, fixed before the results: "Draft a pre-analysis plan: horizons, out-of-sample start date, benchmark, test statistic and the multiple-testing correction across horizons."
 - ▶ Placebo: VRP lagged 12 months or block-shuffled; subsamples; 5-minute realised variance of SPY (from 2020) instead of daily squared returns
- **Critique**: "Act as a hostile referee: why could this out-of-sample R^2 be spurious or biased, and which test settles each objection?"
 - ▶ Expect: bias with persistent regressors, overlap, the choice of start date, the weight of 2008 and 2020
- **Report**: every horizon and specification tried, in one table; the plan dated before the results; AI use logged

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